

TradingAgents

LLM-Augmented Intraday Price Spike Classification

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Problem Statement

When a stock price spikes sharply within a 5-minute window during regular trading hours, will it **continue** in the same direction or **reverse** over the next 60 minutes?

This is a three-class prediction problem:

- **Continuation** — the spike was driven by real information; price holds or extends.
- **Reversal** — the spike was mechanical (panic, stop cascade, liquidity); price mean-reverts.
- **Unclear** — no decisive 60-minute outcome.

Motivation. Large intraday spikes are precisely where the informed-vs-uninformed trading distinction matters most. Pure price statistics are ambiguous — the same spike pattern can mean fundamentally different things depending on news context, macro regime, and order flow. This motivates bringing in AI agents capable of reasoning over heterogeneous inputs.

Data

OHLCV price data. 5-minute bars from Polygon.io for 11 tickers (AAPL, MSFT, NVDA, TSLA, META, AMD, NFLX, PLTR, SPY, QQQ, XLK), covering October 2025 – January 2026 (≈ 84 trading days). Total: 126,719 rows stored in `ohlcv_5min.parquet`.

News data. For each selected event, the Polygon.io news API is queried for articles published within a 3-hour pre-event window. Per-event JSON packets stored in `out_news/packets/`.

Label proxy. The 60-minute forward return from t_0 determines the ground-truth label: $>0.3\%$ same direction $\rightarrow$ continuation; $>0.3\%$ opposite $\rightarrow$ reversal; otherwise unclear. Labels are *never* shown to agents.

Pipeline Overview

Step	Script	Output	Count
1	<code>mine_events_strict.py</code>	Candidate events	451
2	<code>select_best_events.py</code>	Balanced event set	100
3	<code>build_news_packets.py</code>	News JSON per event	100
4	<code>run_ip_v3.py</code>	Agent outputs (JSONL)	100
5	<code>eval/ml_stage.py</code>	Final predictions	85 (clear)

Event mining uses a two-stage filter: Stage 1 flags bars with return z-score ≥ 3 , volume $\geq 4\times$, or range $\geq 3\times$ typical. Stage 2 confirms with stricter thresholds ($z \geq 4$, absolute move $\geq 60\text{bp}$) plus a persistence check (6-bar return $\geq 50\%$ of 3-bar peak).

Event selection enforces per-ticker quotas (8–15 events for stocks, 3–8 for ETFs), caps unclear events at 15%, and applies a 60-minute temporal de-duplication window per ticker to avoid correlated events.

Agent Architecture — Version History

V1 and V2.1 — Multi-Agent Debate (5 LLM calls/event)

The initial design framed each spike as a mixture of **Information** (I) and **Panic** (P) components: $I > P \rightarrow \text{continuation}$; $P > I \rightarrow \text{reversal}$. Three specialist agents (Microstructure, News, Macro) each produced I/P scores. A Skeptic agent critiqued them; agents revised; a Judge issued the final verdict.

V1 result: 35.2%. The Judge exhibited strong reversal bias.

V2.1 result: 16.3%. Worse than V1 due to *Degeneration-of-Thought* [1]: the Microstructure agent’s confident continuation scores “infected” all other agents through the Skeptic round. The Judge predicted continuation 95/98 times; true continuations are only 43% of events, yielding 16.3%.

V3 — Direct-Label Ensemble (3 LLM calls/event)

Key design changes:

- Agents output *direct labels* (continuation/reversal) not I/P scores.
- Skeptic and Judge eliminated; ensemble is pure mathematics.
- ReTuning evidence-scoring format [3]: agents enumerate evidence *for each label* (0–20 points) before committing.
- “Unclear” is forbidden per-agent — only the ensemble can declare it.
- 7 new pre-event microstructure features added.

Model routing. GPT-4o-mini (OpenAI) for Microstructure; Claude Haiku (Anthropic) for News; DeepSeek-Chat for Macro. Intentional provider diversity reduces correlated errors [2].

Agent inputs:

Agent	Receives
Micro	Spike stats (z-score, volume mult, cluster length) + 7 V3 microstructure features
News	Up to 5 pre-event articles (strictly before t_0) + publication timestamps
Macro	SPY/QQQ 15-min and 60-min pre-event returns + up to 4 macro headlines

The 7 V3 Microstructure Features

Feature	Finance meaning
<code>idio_resid_bp</code>	Stock return $-\beta \cdot \text{SPY}$ return. Large value $\rightarrow$ stock-specific overreaction $\rightarrow$ reversal.
<code>rs_ratio</code>	Downward semivariance / total variance in pre-event bars. $\approx 0.5 \rightarrow$ noisy two-sided action $\rightarrow$ reversal.
<code>signed_vol_ratio_pre</code>	Net buyer/seller pressure in 5 bars before t_0 . Strong one-side $\rightarrow$ informed $\rightarrow$ continuation.
<code>vpin_proxy_pre</code>	Order flow imbalance (pre-event). High VPIN $\rightarrow$ informed trader $\rightarrow$ continuation.
<code>spread_bp</code>	Estimated bid-ask spread at t_0 . Wide spread $\rightarrow$ market-maker uncertainty $\rightarrow$ reversal.
<code>pre_spike_run_len</code>	Consecutive bars in spike direction before t_0 . Long run $\rightarrow$ exhaustion $\rightarrow$ reversal.
<code>is_macro_driven</code>	$ \text{SPY z-score} > 1.5$ at t_0 . Market-wide move $\rightarrow$ efficient absorption $\rightarrow$ continuation.

Weighted Math Ensemble

After agents vote, a direction heuristic acts as a 4th voter (up $\rightarrow$ reversal, down $\rightarrow$ continuation), justified by the intraday mean-reversion literature (Jegadeesh 1990, Lehmann 1990):

$$\text{votes}[\ell] = w_{\text{micro}} \cdot c_{\text{micro}} + w_{\text{news}} \cdot c_{\text{news}} + w_{\text{macro}} \cdot c_{\text{macro}} + w_{\text{dir}} \cdot p_{\text{dir}}$$

where c_i is agent confidence and $w = \{0.35, 0.20, 0.15, 0.22-0.38\}$. The direction weight varies by time-of-day (higher in the last 2 hours; lower at open). If $\max(\text{votes}) < 0.35$ the ensemble returns “unclear.”

The reversal prior p_{dir} is asymmetric: 0.62 for $z > 3$ (large spikes revert more), reduced by 15% if macro-driven, and reduced by 18% for down-spikes (leverage asymmetry).

ML Stage — LightGBM + RF + ExtraTrees

V3 agent outputs (labels, confidence scores, score ratios) are combined with the 7 microstructure features and event statistics into a **35-feature matrix** over 85 clear-label events (100 minus 15 unclear).

Three classifiers are trained with **Leave-One-Out cross-validation** (LOO-CV): on each fold, 84 events train and 1 tests. LOO-CV is the most conservative estimator for small N .

$$\text{Stacked prob} = \frac{1}{3} p_{\text{LGBM}} + \frac{1}{3} p_{\text{RF}} + \frac{1}{3} p_{\text{ET}}$$

Classifier settings. LightGBM: `num_leaves=8`, `min_child_samples=5`, `lr=0.05`. Random-Forest and ExtraTrees: `max_depth=4`, `min_samples_leaf=4`, `class_weight=balanced`.

Data Leakage — Discovery and Fix

The bug. An external reviewer found that the top 3 LightGBM features (`vpin_proxy`, `cluster_len`, `signed_vol_ratio`) were computed using bars *starting at or after* t_0 , not strictly before. The original code:

```
at_after = sub[sub["timestamp"] >= t0]    # >= includes t0 and beyond
return at_after.iloc[:cluster_len]        # post-event bars
```

Because these bars overlap with the outcome window, the model partially learned “given early outcome, predict outcome” — a form of target leakage.

The fix.

- `vpin_proxy` and `signed_vol_ratio` recomputed from the 5 bars strictly before t_0 .
- `cluster_len` removed from the ML feature matrix entirely.
- `strength_score` removed (contained `cluster_len` at 20% weight).

Impact:

Stacked accuracy	
Before fix (leaky)	82.4%
After fix (clean)	67.1%

The 15.3pp drop is entirely attributable to the removed features. 67.1% is the honest number.

Remaining disclosures. `cluster_len` still appears in agent briefs (indirect, unquantifiable contamination of LLM outputs; would require re-running all LLM calls to fix). Event selection used `label_proxy` to prioritize clear-outcome events — this is a sample selection bias, not model leakage, but limits generalizability.

Results

Strategy	3-way (N=100)	Clear subset (N=85)
V2.1 full pipeline	16.3%	9.6%
V1 full pipeline	35.2%	—
Always reversal	42.0%	49.4%
Always continuation	43.0%	50.6%
Direction heuristic	48.0%	56.5%
V3b ensemble	49.0%	57.6%
Math-only LGBM (no agents)	—	57.6%
Agents-only LGBM (no microstr.)	—	61.2%
Full LightGBM LOO-CV	—	62.4%
Full RandomForest LOO-CV	—	65.9%
Stacked LGBM+RF+ET	—	67.1%

Top LightGBM features (by mean LOO-fold importance): `vpin_proxy_pre` (114), `cluster_len*` (92, removed), `signed_vol_ratio_pre` (88), `peak_range_mult` (38), `idio_resid_bp` (35). Microstructure features dominate; LLM agent labels have secondary importance.

Ablation Study

Feature set	LGBM LOO-CV	vs. baseline
Math-only (no LLM features)	57.6%	+1.1pp vs heuristic
Agents-only (no microstructure)	61.2%	+4.7pp vs math-only
Full stack	62.4%	+4.8pp vs math-only

Finding. LLM agents contribute a genuine +4.7pp (“Contextual Alpha”) over pure microstructure math. This confirms that agents add signal through news interpretation and macro reasoning beyond what numbers alone capture. However, the primary signal source is microstructure — agents are secondary.

Constants Stability

The V3 ensemble has 14 hardcoded constants (weights, priors, thresholds). We perturbed each by $\pm 10\%$ and $\pm 20\%$ and measured accuracy change on stored agent outputs (no LLM re-runs needed).

Constant	Max $ \Delta $ at $\pm 10\%$	Max $ \Delta $ at $\pm 20\%$
<code>unclear_thresh</code> (0.35)	2.0pp	3.1pp
<code>w_micro</code> (0.35)	1.0pp	2.0pp
All other 12 constants	0.0–1.0pp	0.0–2.0pp

9 of 14 constants have zero effect at $\pm 10\%$. The system is not tuned to the 85 events.

Limitations and Future Work

1. **Small N .** $N=85$ is the primary weakness. LOO-CV is conservative but external validation on a held-out time period is required before deployment claims.
2. **Short data window.** Only ≈ 3 months of data (Oct 2025 – Jan 2026). Extending to 1+ FY would provide more diverse market regimes.
3. **Residual brief contamination.** `cluster_len` is shown in agent briefs but removed from ML features. Its indirect influence on agent outputs is unquantifiable without re-running all LLM calls.
4. **Sample selection bias.** Events selected partly for clear outcomes — results may overstate performance on ambiguous real-world events.
5. **Future directions.** Expand tickers and time window; add options flow and earnings calendar features; test whether agent Δ accuracy holds out-of-sample.

References

- [1] Liang et al. (2024). *Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate*. EMNLP 2024. arXiv:2305.19118
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- [5] Jegadeesh, N. (1990). *Evidence of Predictable Behavior of Security Returns*. Journal of Finance.
- [6] Lehmann, B. (1990). *Fads, Martingales, and Market Efficiency*. Quarterly Journal of Economics.